ModelIt!

Teacher's Guide

# Rock Cycle Deep Dive: From Magma to Mountain

How Earth Recycles Itself Over Millions of Years

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 07L2 |
| **NGSS Standard** | MS-ESS2-1, MS-ESS1-4 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: MS-ESS2-1, MS-ESS1-4**

Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process. Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's 4.6-billion-year-old history.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students develop a model to describe the cycling of Earth's materials through igneous, sedimentary, and metamorphic rock formation. |
| **Disciplinary Core Idea** | **ESS2.A Earth's Materials and Systems / ESS1.C The History of Planet Earth** — All Earth processes are the result of energy flowing and matter cycling. Rock strata provide evidence for the geologic time scale. |
| **Crosscutting Concept** | **Energy and Matter** — Students trace the flow of energy (internal heat, solar) that drives matter cycling through the rock cycle, showing how energy inputs determine which transformations occur. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A geologist finds a rock that was once sandstone (sedimentary) but now has a crystalline structure with folded layers. Which process most likely caused this transformation and what type of rock is it now?

**CONSTRUCTED RESPONSE:**

Using your rock cycle model, explain how a single piece of granite (igneous rock) could eventually become sandstone (sedimentary rock) and then marble (metamorphic rock). Describe the specific processes and conditions required at each step.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The rock cycle is Earth's grand recycling system. Over billions of years, the planet has continuously created, destroyed, and recreated rocks through processes driven by internal heat and external energy from the Sun. Understanding the rock cycle explains how Earth's surface has changed over its 4.6-billion-year history.

## Three Rock Types

Igneous rocks form when magma (underground) or lava (surface) cools and solidifies. Granite forms slowly underground (large crystals); basalt forms quickly on the surface (small crystals). Sedimentary rocks form when weathered fragments are deposited, compacted, and cemented. Sandstone, limestone, and shale are common examples. Metamorphic rocks form when existing rocks are transformed by intense heat and pressure without melting. Marble comes from limestone; slate comes from shale.

## The Cycling Process

The rock cycle has no starting point. Igneous rock at the surface weathers into sediment, which becomes sedimentary rock. If buried deep, sedimentary rock becomes metamorphic. If heated further, it melts into magma, which cools into new igneous rock. But shortcuts exist: igneous rock can become metamorphic directly, or metamorphic rock can weather directly into sediment.

## Energy Driving the Cycle

Two energy sources power the rock cycle. Earth's internal heat (from radioactive decay and residual formation heat) drives melting, metamorphism, and plate tectonics. The Sun's energy drives weathering and erosion by powering wind, water cycles, and temperature changes. Without both energy sources, the rock cycle would stop.

## Rock Strata and Geologic Time

Sedimentary rock layers (strata) are deposited in sequence, with older layers on the bottom and younger layers on top (Law of Superposition). Fossils in these layers record the history of life. By studying rock strata, scientists have organized Earth's 4.6-billion-year history into eons, eras, periods, and epochs, creating the geologic time scale.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify heat and pressure, weathering rate, igneous rock formation, sedimentary layers, and metamorphic transformation as components of the rock cycle system.

**E** **Establish Relationships**
Students determine that heat/pressure drives metamorphic and igneous formation while weathering drives sedimentary formation, with the cycle connecting all three types.

**V** **Visualize & Model**
Students build a model showing how the rock cycle continuously transforms Earth's materials based on conditions at different depths.

**E** **Evaluate Outcomes**
Students run deep underground, surface erosion, and balanced scenarios to observe how different conditions favor different rock types.

**R** **Revise & Extend**
Students add tectonic plate movement or fossil preservation to explore how plate tectonics connects the deep and surface rock processes.

### Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Pick up any rock. It might be older than the dinosaurs. Or it might have been magma last year. How can you tell?"*

**Watch For**

Pass around three rock samples (granite, sandstone, marble). Ask: What's different about these?

## Slide 2: Learning Objectives

3 min

**Say:** *"Today we're diving deep into how Earth recycles itself over billions of years."*

**Watch For**

Have students read objectives. Pre-teach 'igneous,' 'sedimentary,' and 'metamorphic.'

## Slide 3: Big Question

3 min

**Say:** *"Earth is 4.6 billion years old. But some rocks formed last week from volcanic eruptions. How is the planet constantly making new rocks?"*

**Watch For**

Show before/after images of volcanic eruptions creating new land. Discuss.

## Slide 4: LEVER Framework

2 min

**Say:** *"We'll build a model of Earth's rock recycling system."*

**Watch For**

Preview LEVER steps. Show rock cycle diagram for reference.

## Slide 5: Activity 1: Components

8 min

**Say:** *"What forces shape rocks? What can we control in our model vs. what responds?"*

**Watch For**

Guide sorting. Discuss why heat/pressure and weathering are external driving forces.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"When heat and pressure increase, which rock types form? When weathering increases, what builds up?"*

**Watch For**

Help students trace connections. Emphasize that the cycle has NO starting point.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Let's crank up the heat underground and see what happens. Then let's crank up weathering on the surface."*

**Watch For**

Students predict first, then run all three scenarios. Compare which rock types dominate.

## Slide 8: Discoveries

5 min

**Say:** *"So WHY is the rock cycle a cycle and not a chain? What did our model prove about Earth's recycling?"*

**Watch For**

Lead discussion about how the cycle maintains all three rock types. Connect to geologic time.

## Slide 9: STEM Challenge

12 min

**Say:** *"You're geologists now. Tell the story of 4.6 billion years using rocks as your evidence."*

**Watch For**

Groups create geologic timelines with rock type examples at key events.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Heat & Pressure (Depth), Weathering Rate
- **INTERNAL:** Igneous Rock Formation, Sedimentary Layers, Metamorphic Transformation

## Relationships

- **Heat & Pressure to Metamorphic Transformation:** POSITIVE (+) — Greater heat and pressure deep underground cause more rock to undergo metamorphic transformation, changing its mineral structure without melting it.
- **Heat & Pressure to Igneous Rock Formation:** POSITIVE (+) — Extreme heat melts rock into magma. When magma cools (either underground or at the surface), it solidifies into igneous rock.
- **Weathering Rate to Sedimentary Layers:** POSITIVE (+) — Faster weathering breaks down more surface rock into sediment. This sediment is transported, deposited, and compressed into sedimentary rock layers.
- **Sedimentary Layers to Metamorphic Transformation:** POSITIVE (+) — As sedimentary layers accumulate, the weight of upper layers creates pressure on deeper layers. If buried deep enough, this pressure (combined with heat) triggers metamorphic transformation.
- **Weathering Rate to Igneous Rock Formation:** NEGATIVE (-) — High weathering erodes and breaks down exposed igneous rock at the surface, reducing the amount of intact igneous rock and converting it to sediment.

## Key Discoveries

- The rock cycle has no beginning or end: any rock type can become any other rock type given the right conditions
- Heat and pressure deep underground create igneous and metamorphic rocks, while weathering at the surface creates sedimentary rocks
- The rock cycle is driven by Earth's internal heat (radioactive decay and leftover formation heat) and external energy (the Sun, which drives weathering)
- Sedimentary rock layers contain fossils and provide a timeline of Earth's history, which is how scientists organize the geologic time scale

### Multiple Choice Post-Assessment Answer Key

**1. A geologist finds a rock that was originally sandstone (sedimentary) but now has a crystalline, folded structure. Based on the rock cycle model, which process and resulting rock type best explains this transformation?**

**Correct: B) The sandstone was buried deep enough to be subjected to intense heat and pressure, transforming it into metamorphic rock without melting**

*Correct! When sedimentary rock is buried deeply enough, increasing heat and pressure transform its mineral structure without melting it. This metamorphic transformation changes sandstone into quartzite, which displays the crystalline, folded structure the geologist observed.*

**2. In the rock cycle model, Sedimentary Layers have a positive relationship with Metamorphic Transformation. Which mechanism explains why the accumulation of sedimentary layers can lead to metamorphism?**

**Correct: B) As sedimentary layers accumulate, the weight of upper layers creates increasing pressure on deeper layers. Combined with geothermal heat at depth, this triggers metamorphic transformation**

*Correct! As sedimentary layers stack up over millions of years, the weight of the overlying material creates enormous pressure on the deepest layers. Combined with increasing geothermal heat at greater depths, this pressure is sufficient to transform the mineral structure of the rock without melting it.*

**3. The model shows that the rock cycle has no beginning or end, meaning any rock type can become any other rock type. A student argues that sedimentary rock must always become metamorphic before becoming igneous. Using the model, which scenario disproves this claim?**

**Correct: A) Sedimentary rock can be subducted directly into the mantle where extreme heat melts it into magma, which cools into igneous rock, bypassing the metamorphic stage entirely**

*Correct! The rock cycle has multiple pathways, not a single sequence. Sedimentary rock can be pulled into the mantle at subduction zones where temperatures are high enough to melt it directly into magma. When that magma cools, it forms igneous rock without ever passing through a metamorphic stage.*

**4. In the model, when Heat & Pressure is set to low and Weathering Rate is set to high, Sedimentary Layers dominate while Igneous Rock Formation and Metamorphic Transformation are minimal. If Earth's internal heat source completely shut off, what would the model predict for the long-term rock cycle?**

**Correct: B) Only weathering and sedimentary formation would continue because the Sun drives surface erosion, while metamorphic and igneous processes would stop without internal heat**

*Correct! The rock cycle is powered by TWO energy sources: internal heat (driving metamorphism, melting, and plate tectonics) and solar energy (driving weathering and erosion). Without internal heat, only surface processes would continue. Over time, all exposed rocks would weather into sediment. Mars is an example of a planet where this has happened.*

**5. Geologists use the Law of Superposition to determine relative rock ages: in undisturbed strata, older layers are on the bottom and younger layers are on top. A scientist discovers a fossil of a marine organism embedded in rock near a mountain summit. Using the rock cycle model, which sequence of events best explains this finding?**

**Correct: B) Marine sediment containing the organism was deposited on the ocean floor, lithified into sedimentary rock, then tectonic forces uplifted the rock from the seafloor to mountain height over millions of years**

*Correct! The rock cycle and plate tectonics work together. Marine organisms die and settle on the ocean floor where they are buried in sediment and fossilized in sedimentary rock. Over millions of years, tectonic forces can push these seafloor rocks upward, even creating mountain ranges. This is why marine fossils are found at high elevations worldwide.*

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Rocks don't change" | Rocks are constantly changing, just very slowly from a human perspective. Over millions of years, mountains erode to plains, seafloors become mountaintops, and solid rock melts to magma and solidifies again. The rock cycle transforms rocks continuously. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "The rock cycle goes in one direction" | The rock cycle has NO fixed direction. Any rock type can become any other type. Igneous can become sedimentary OR metamorphic. Sedimentary can become metamorphic OR melt to become igneous. The path depends entirely on the conditions the rock encounters. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Fossils are only found in very old rocks" | Fossils are found in sedimentary rocks of ALL ages, from billions of years old to very recent. Fossils are forming right now as organisms die and are buried in sediment. What changes over time is the TYPE of organisms preserved: ancient rocks contain simple organisms, while recent rocks contain more complex life. |